

MIT OpenCourseWare

2.25

Advanced Fluid Mechanics 2.25 — Advanced Fluid Mechanics May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Construct an original proof-level problem involving Navier-Stokes for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
2. (3 points) Analyze the limitations of standard approaches to Vorticity Dynamics when applied to edge cases in 2.25. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (4 points) Prove the fundamental theorem concerning Turbulence in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
4. (8 points) Prove the fundamental theorem concerning Instability in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (4 points) Construct an original proof-level problem involving Boundary Layers for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (4 points) Prove the fundamental theorem concerning Rheology in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
7. (3 points) Prove the fundamental theorem concerning Navier-Stokes in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (3 points) Prove the fundamental theorem concerning Vorticity Dynamics in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (3 points) Prove the fundamental theorem concerning Turbulence in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
10. (9 points) Prove the fundamental theorem concerning Instability in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
11. (2 points) Prove the fundamental theorem concerning Boundary Layers in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (3 points) Prove the fundamental theorem concerning Rheology in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
13. (3 points) Prove the fundamental theorem concerning Navier-Stokes in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
14. (5 points) Analyze the limitations of standard approaches to Vorticity Dynamics when applied to edge cases in 2.25. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (2 points) Prove the fundamental theorem concerning Turbulence in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (2 points) Construct an original proof-level problem involving Instability for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
17. (5 points) Construct an original proof-level problem involving Boundary Layers for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
18. (5 points) Construct an original proof-level problem involving Rheology for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
19. (3 points) Prove the fundamental theorem concerning Navier-Stokes in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

20. (7 points) Construct an original proof-level problem involving Vorticity Dynamics for 2.25 (Advanced Fluid Mechanics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
21. (5 points) Prove the fundamental theorem concerning Turbulence in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
22. (4 points) Prove the fundamental theorem concerning Instability in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Prove the fundamental theorem concerning Boundary Layers in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (3 points) Prove the fundamental theorem concerning Rheology in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
25. (3 points) Prove the fundamental theorem concerning Navier-Stokes in the context of 2.25 (Advanced Fluid Mechanics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Navier-Stokes from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Vorticity Dynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 4:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Boundary Layers from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 7:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 10:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 11:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 13:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 14:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Vorticity Dynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

Solution approach: First recall the definitions and key theorems related to Instability from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 17:

Solution approach: First recall the definitions and key theorems related to Boundary Layers from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:

Solution approach: First recall the definitions and key theorems related to Rheology from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 19:

The proof follows from the definitions and properties established in 2.25. The key

insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 20:

Solution approach: First recall the definitions and key theorems related to Vorticity Dynamics from 2.25. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 21:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 22:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 25:

The proof follows from the definitions and properties established in 2.25. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.